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Diastereodivergent Synthesis of β-Amino Alcohols by Dual-Metal-Catalyzed Coupling of Alkoxyallenes with Aldimine Esters

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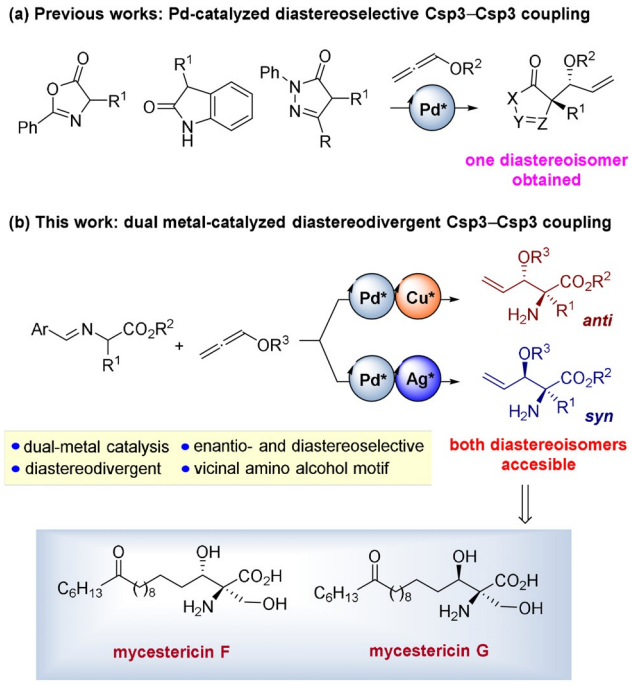
Dedicated to the 70th anniversary of Shanghai Institute of Organic Chemistry (SIOC)

Abstract: *Both syn- and anti-β-amino alcohols are common structural motifs in natural products, drug molecules, chiral ligands and catalysts. However, the currently available methods for synthesizing these motifs are limited to generate only one diastereoisomer. Therefore, development of a unified method for stereoselective access to complementary diastereomers would be highly desirable. Herein, we report a method for dual-metal-catalyzed diastereodivergent coupling of alkoxyallenes with aldimine esters. By carefully selecting the two metals and appropriate chiral ligands, we could synthesize both syn- and anti-β-amino alcohol motifs with high enantioselectivity and diastereoselectivity from the same set of starting materials. Furthermore, stereodivergent syntheses of all four stereoisomers of β-amino alcohols could be achieved. We demonstrated the synthetic utility of this method by concisely synthesizing two β-amino alcohol natural products, mycestericins F and G.*

Introduction

β-Amino alcohol structural motifs are present in more than 300,000 compounds, including >2,000 natural products, >80 FDA-approved drugs and >100 drug candidates.[1] β-Amino alcohols are also important building blocks for chiral ligands and auxiliaries used in asymmetric synthesis.[2] Compounds with this moiety are conventionally prepared by means of multistep transformations of α-amino acid derivatives[3] or by stereoselective oxyamination of alkenes,[4] but catalytic Csp^3-Csp^3 coupling reactions have emerged as a powerful alternative, owing to their atom- and step-economy. However, only a few of the reported coupling reactions are suitable for preparing β-amino alcohols with vicinal stereocenters.[5,6] For example, Malcolmson et al. reported elegant reductive coupling reactions of azadienes and ketones to access *anti*-β-amino tertiary alcohols.[5a] Krische and co-workers enantioselectively synthesized *anti*-β-amino alcohols by reductive coupling reactions of phthalimidoallenes and aldehydes, which proceed via a hydrogen auto-transfer process.[5b] Despite these achievements, none of the existing methods provide access to both *syn*- and *anti*-β-amino alcohols. Diastereodivergent access to complementary diastereomers of β-amino alcohols, especially from the same set of starting materials, would be highly desirable because both *syn*- and *anti*-β-amino alcohols are useful for organic synthesis.

Transition-metal-hydride-catalyzed coupling reactions of unsaturated hydrocarbons and enols or enolates has proven to be a reliable method for asymmetric construction of C–C bonds.[7–9] In particular, alkoxyallenes can be used as coupling partners for atom-economical preparation of chiral allylic alcohol derivatives.[10] In pioneering work, Trost and co-workers reported highly enantio- and diastereoselective additions of azlactones and oxindoles to alkoxyallenes (Scheme 1 a).[10a,b] More recently, Jiang and co-workers developed a protocol for Pd-catalyzed or Brønsted acid catalyzed addition reactions of pyrazolones and alkoxyallenes with divergent regioselectivity.[10c] In these reactions, single-catalyst systems are employed, and the diastereoselectivity is controlled by catalyst-substrate and substrate-substrate interactions. Therefore, only the kinetically favored diastereoisomer is accessible. Stereodivergent transition-metal-hydride-catalyzed coupling reactions remain a formidable chal-



Scheme 1. Pd-hydride-catalyzed coupling reactions of alkoxyallenes and carbon nucleophiles.

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lenge.[11] Rare examples include Dong's work on cooperative Rh-H/amine-catalyzed coupling of aldehydes with alkynes[11a] and our recent contribution involving synergistic Pd-H/Cu-catalyzed coupling of 1,3-dienes with aldimine esters.[11b] In these synergistic catalyst systems,[12] the two chiral catalysts independently control the stereochemistry of the nucleophile and electrophile during the coupling step; therefore, stereodivergent synthesis[13] can be achieved simply by choosing the appropriate combination of catalyst chiralities.

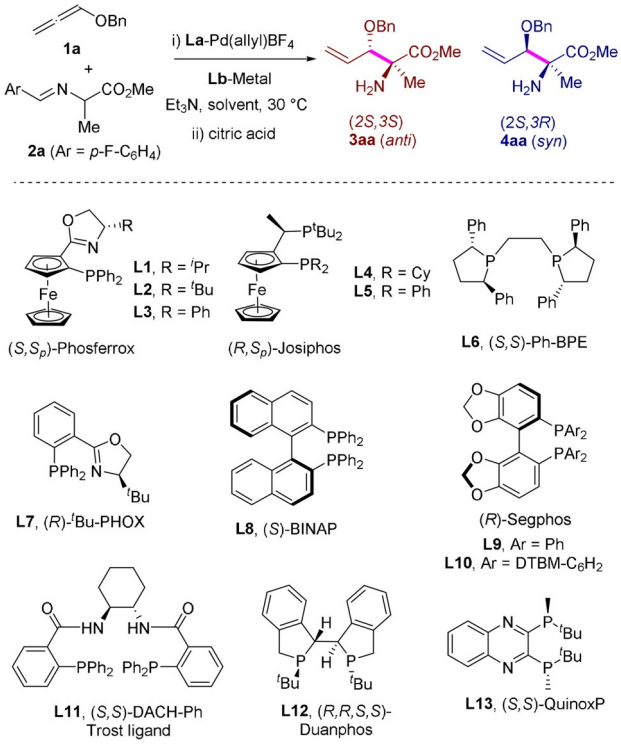
Inspired by the aforementioned advances as well as the recent achievements in stereodivergent allylic alkylation reactions,[14–16] herein we report a protocol for dual-metal-catalyzed diastereodivergent coupling of alkoxyallenes with aldimine esters as an efficient method for preparing both *syn*- and *anti*-β-amino alcohol motifs with high enantio- and diastereoselectivities (Scheme 1b). In addition, all four stereoisomers of β-amino alcohols could be stereodivergently synthesized simply by choosing the appropriate combination of catalyst chiralities. Notably, after hydroxylation of the ester group of the products to a carboxylic acid, this method provided a stereodivergent access to β-hydroxy α-amino acids, which are one important type of unnatural α-amino acids occasionally being observed in many bioactive natural products. Previous works towards synthesis of these motifs mainly relied on aldol type reactions, which could only obtain one diastereisomer[17] and a biocatalyisis pathway to access all four stereoisomers were reported but it gave moderate diastereoselectivities.[18] Therefore, the dual-metal catalysis strategy used herein showed huge superiority over the traditional aldol reactions. Finally, as a demonstration of this method, we used the coupling reaction as a key step for the total syntheses of two β-amino alcohol natural products, mycestericins F and G.

Results and Discussion

We began by investigating a Cu/Pd catalyst system[11b,19,20] for the coupling of alkoxyallene **1a** with aldimine ester **2a** as model substrates (Table 1). With (S,S_p)-Phosferrox **L1** as the chiral ligand for Cu, we tested the following bidentate chiral ligands for Pd: (R,S_p)-Josiphos **L4** and **L5**, (*S*,*S*)-Ph-BPE (**L6**), (*R*)-^{t}Bu-PHOX (**L7**), (*S*)-BINAP (**L8**), (*R*)-Segphos **L9** and **L10**, and (*S*,*S*)-DACH-Ph Trost ligand (**L11**) (entries 1–8). Although good to excellent enantioselectivities were obtained, the diastereoselectivities were low, as indicated by the **3aa**/**4aa** ratios. Screening of additional ligands revealed that C2-symmetric bisphosphine ligands with phosphorus chirality gave good stereocontrol. For example, a 7:1 **3aa**/**4aa** ratio was obtained by using (*R*,*R*,*S*,*S*)-DuanPhos (**L12**) as the Pd ligand (entry 9), and an even higher ratio was achieved with (*S*,*S*)-QuinoxP **(L13)**: *anti*-selective coupling product (*2S*,*3S*)-**3aa** was obtained in 71% isolated yield with >99% *ee* and 14:1 *anti*/*syn* selectivity (entry 10).

Next, we attempted to prepare the more challenging *syn*-selective coupling product.[21] We first tested the combination of **L1**-Cu with *ent*-**L13**-Pd (Table 1, entry 11). Although the *anti*-coupling product did indeed predominate under these conditions, the *syn*/*anti* ratio was only moderate, and the yield dropped to 64%. We speculated that there may have been strong noncovalent interactions between the Cu and Pd catalysts during the coupling step and thus that mismatched

Table 1: Optimization of conditions for the coupling reaction.[a]



Entry	La	Lb-M	Solvent	3aa/4aa[b]	Yield [%][b]	*ee* [%][c]
1	**L4**	**L1-Cu**	THF	2:1	57	98/97
2	**L5**	**L1-Cu**	THF	1.4:1	63	99/97
3	**L6**	**L1-Cu**	THF	3.7:1	66	>99/99
4	**L7**	**L1-Cu**	THF	1:2.2	54	97/98
5	**L8**	**L1-Cu**	THF	1:1.6	70	81/97
6	**L9**	**L1-Cu**	THF	1.1:1	69	82/98
7	**L10**	**L1-Cu**	THF	1:2	65	>99/97
8	**L11**	**L1-Cu**	THF	1:4.7	65	>99/>99
9	**L12**	**L1-Cu**	THF	7:1	78	>99/>99
10	**L13**	**L1-Cu**	THF	14:1	80(70[e])	>99/nd
11	*ent*-**L13**	**L1-Cu**	THF	1:3.9	64	>99/>99
12	**L1**	**L1-Cu**	THF	1:4.1	73	>99/>99
13	**L2**	**L2-Cu**	THF	1:4.5	64	99/>99
14	**L3**	**L3-Cu**	THF	1:5.2	60	>99/>99
15	**L3**	**L3-Ag**	THF	1:12	42	nd/>99
16	**L3**	**L3-Ag**	PhMe	1:14	72	nd/>99
17	**L3**	**L3-Ag**	DCM	1:2.5	54	41/90
18	**L3**	**L3-Ag**	dioxane	1:14	48	nd/>99
19	**L3**	**L3-Ag**	CPME	1:18	80	nd/>99
20[d]	**L3**	**L3-Ag**	CPME	1:20	82(73[e])	nd/>99
21	**L13**	–	THF	nd	<5	nd/nd
22	**L3**	–	CPME	nd	<5	nd/nd

[a] Reaction conditions: (i) **1a** (0.1 mmol), **2a** (0.2 mmol), **La**-Pd(allyl)BF_4 (4 mol%), **Lb** (6 mol%), **M** (5 mol%), Et_3N (200 mol%), solvent (0.5 mL), 30°C, 48 h. (ii) Aqueous citric acid (10%), 1 h. Cu = $Cu(CH_3CN)_4PF_6$, Ag = AgOAc. The regioselectivity was >20:1 in all cases. THF, tetrahydrofuran; DCM, dichloromethane; CPME, cyclopentyl methyl ether. nd = not determined. [b] The *anti*/*syn* (**3aa**/**4aa**) ratios and yields were determined from the ^{1}H NMR spectra of the crude products; nd, not determined. [c] The *ee* values were determined by chiral HPLC. [d] Reaction with 5 mol% **L3** and 6 mol% AgOAc. [e] Isolated yield.

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catalyst chirality may have led to the poor diastereoselectivity. After exploring a series of Pd and Cu catalyst combinations, we discovered that *syn*-coupling to afford (*2S,3R*)-**4aa** predominated when the same (S,S_p)-Phosferrox ligand was used for both Pd and Cu (entries 12–14). Among the tested ligands, phenyl-substituted (S,S_p)-Phosferrox (**L3**) gave the best *syn*/*anti* ratio (5.2:1, entry 14).

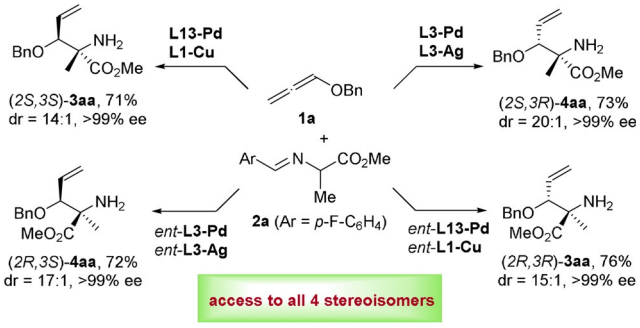
We hypothesized that because Ag^I has a larger ionic radius than Cu^I, its complex with aldimine **1a** would exhibit greater structural flexibility,[22] which would help to reduce ligand-ligand interference and improve control of the stereochemistry.[23] To test this hypothesis, we carried out reactions in which $Cu(CH_3CN)_4PF_6$ was replaced with AgOAc, and we were pleased to find that the **L3-**Ag/**L3-**Pd catalyst system increased the *syn*/*anti* ratio to 12:1, although the yield was slightly low (compare entries 14 and 15 in Table 1). In attempts to improve both the selectivity and the yield, we investigated various solvents (entries 16–19). The best reaction medium was cyclopentyl methyl ether, which gave an 80% yield of *syn*-selective coupling product (*2S,3R*)-**4aa** with >99% *ee* and 18:1 *syn*/*anti* selectivity (entry 19). The yield and *syn*/*anti* ratio were improved further by using a slight excess of AgOAc (entry 20 vs. entry 19).[24] Notably, the regioselectivities of both the Pd/Cu- and Pd/Ag-catalyzed reactions were completely controlled.[13c] As a control experiment, without Cu or Ag catalyst, only utilizing Pd catalyst did not catalyze the reaction (entries 21 and 22).

Having developed conditions for diastereodivergent coupling of alkoxyallenes with aldimine esters, we explored the substrate scope of the reactions, starting with the Cu/Pd-catalyzed *anti*- selective coupling (Table 2). Alkoxyallenes **1** with various R^1 groups were well tolerated in reactions with aldimine ester **2a**. For example, when R^1 was a *para*-methoxybenzyl, *meta*-fluorobenzyl, 2-naphthalenemethyl, 2-indanyl, or phenyl group, corresponding products (2*S*,3*S*)-**3ab**-**3af** were obtained in good yields (54–77%) with good diastereoselectivities (10:1–16:1) and excellent enantioselectivities (96–>99% *ee*). When R^1 was an alkyl group (cyclohexyl, *n*-octyl, or 3-phenylpropyl), the reaction smoothly gave desired products **3ag**–**3ai** with good stereoselectivities. In addition, sterically hindered alkyl alkoxyallenes also underwent the reaction, as demonstrated by the formation of coupling products **3aj** and **3ak**. Alkoxyallenes derived from the natural products (*S*)-prolinol, (*S*)-citronellol, phytol, geraniol, and cholesterol were subjected to the coupling reaction, and found to afford *anti*-β-amino alcohol derivatives **3al**–**3ap**, respectively, in high yields with good diastereoselectivities. When the R^2 group of the aldimine ester was changed from methyl to ethyl, *n*-propyl, or *n*-butyl, slightly lower yields were obtained (**3ba**–**3da**). Similarly, changing the methyl ester to the more bulky *i*-propyl ester also decreased the yield (**3ea**). These results imply that the reaction was sensitive to the steric bulk of the aldimine ester. However, cyclic aldimine esters were well tolerated, giving good yields of products containing a cyclic imine (**3fa**), a lactone (**3ga**), and an oxazoline moiety (**3ha**) with high diastereo- and enantioselectivities. A thioether group was also compatible with the reaction conditions, as shown by the formation of **3ia**, which was obtained in 62% yield with 16:1 dr and >99% *ee*. A ketamine derived from a glycine ester underwent the reaction, giving **3ja** in 56% yield with >99% *ee*, albeit with a poor dr. The stereochemistry of **3ha** was determined by single-crystal X-ray diffraction and the absolute configuration was assigned to (*2S,3S*).[25]

We next examined the scope of the Ag/Pd-catalyzed *syn*-selective coupling reaction (Table 3). A variety of benzyl- and alkyl-substituted alkoxyallenes **1** reacted with **2a** to give the desired *anti*-coupling products (**4ab–4ah**) in good yields with high diastereoselectivities and excellent enantioselectivities. Natural-product-derived alkoxyallenes also showed good reactivities, affording good yields of *syn*-β-amino alcohol derivatives **4ai–4al**. Chiral alkoxyallenes derived from (*R*)- and (*S*)-1-phenylethanol reacted with **2a** to give (*2S,3R,R*)-**4am** and (*2S,3R,S*)-**4an**, respectively. The existing chiral centers in these substrates had little influence on the stereochemistry of the newly formed stereocenters (**4am** vs. **4an**). All the aldimine esters shown in the lower panel of Table 2 were also tested in the Ag/Pd system, and as expected, the corresponding *syn*-coupling products were formed exclusively as the main isomer, in yields ranging from 40 to 85%. In all these examples, the diastereoselectivities were well controlled (dr = 7:1–20:1), except in the case of the reaction of a glycine-ester-derived ketamine (**4ja**), for which the dr was only 2:1. Note also that the enantioselectivity of the Ag/Pd-catalyzed reaction of the methionine-derived aldimine ester was lower than that of the Cu/Pd-catalyzed reaction of the same substrate (80% for **4ia** vs. >99% for **3ia**).

To demonstrate the stereodivergence of this reaction, we coupled **1a** with **2a** with catalysis by four separate catalyst systems: **L13-**Pd/**L1-**Cu, *ent*-**L13-**Pd/*ent*-**L1-**Cu, **L3-**Pd/**L3-**Ag, and *ent*-**L3-**Pd/*ent*-**L3-**Ag (Scheme 2). In this way, we were able to obtain all four stereoisomers of the coupling product-(*2S,3S*)-**3aa**, (*2R,3R*)-**3aa**, (*2R,3S*)-**4aa**, and (*2S,3R*)-**4aa**-in optically pure form (>99% *ee*) with dr values of ≥14:1.

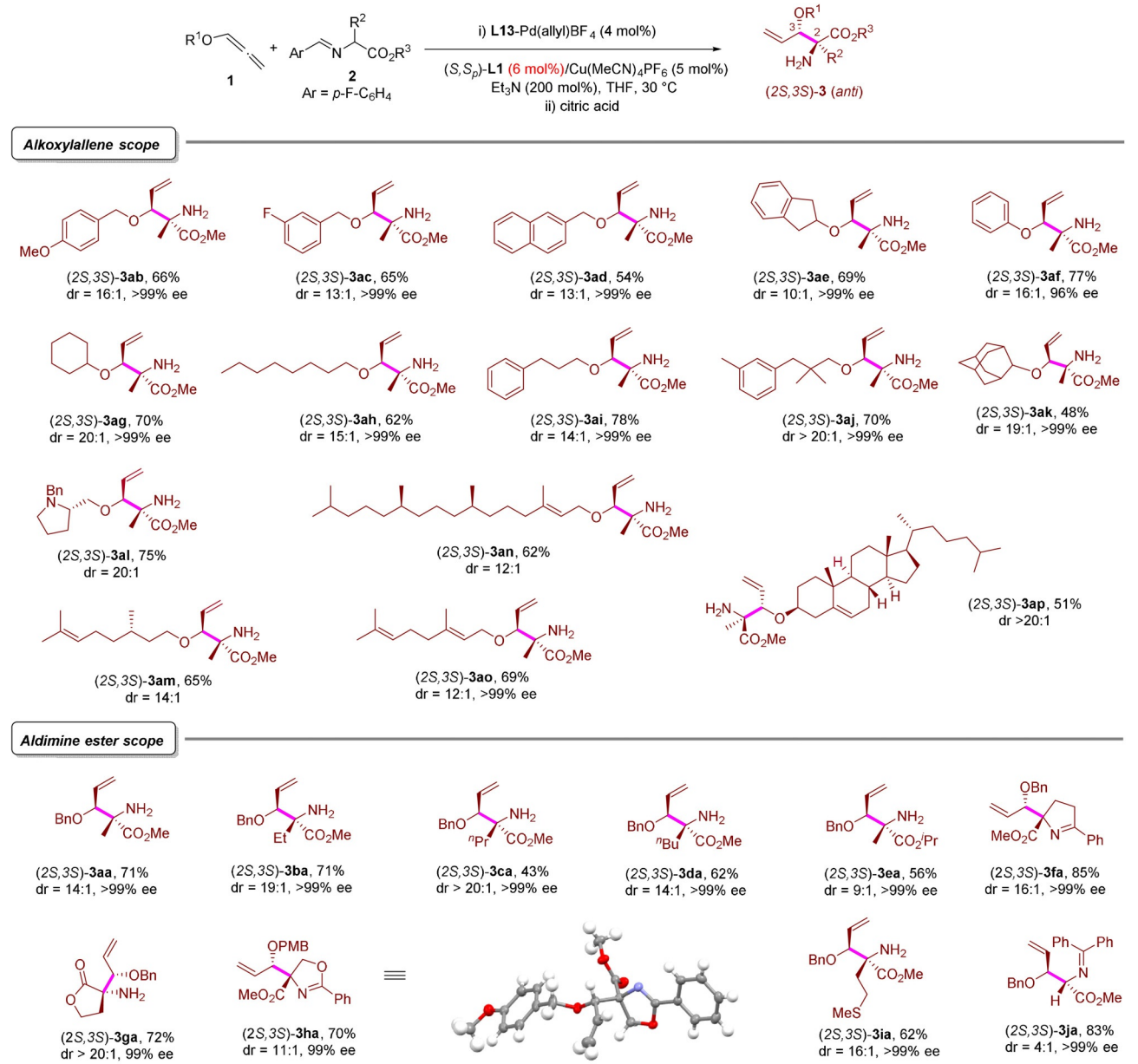
Considerable success in the stereodivergent synthesis of compounds with vicinal stereocenters has been achieved.[13–15] however, stereodivergent preparation of compounds with more than two stereocenters has not yet been reported. Herein, to further demonstrate the robustness of our dual-catalyst systems, we carried out stereodivergent syntheses of amino alcohol derivatives with three stereocenters by means of coupling reactions of alkoxyallenes (*S*)- and (*R*)-**1x** with aldimine ester **2a** (Scheme 3). With appropriate combinations



Scheme 2. Stereodivergent synthesis of products with two stereocenters.

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Table 2: Substrate scope of the *anti*-selective coupling reaction.[a]



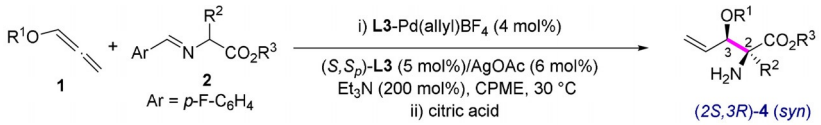
[a] Reaction conditions: (i) **1a** (0.1 mmol), **2a** (0.2 mmol), **L13**-Pd(allyl)BF_4 (4 mol%), **L1** (6 mol%), $Cu(CH_3CN)_4PF_6$ (5 mol%), Et_3N (200 mol%), THF (0.5 mL), 30°C, 48 h. (ii) Aqueous citric acid (10%), 1 h. The regioselectivity was >20:1 in all cases. The dr values were determined from the ^1H NMR spectra of the crude products. Isolated yields are given. The *ee* values were determined by chiral HPLC.

of the substrate enantiomer and the catalyst pair, we were able to obtain good yields of all eight stereoisomers of the coupling products with good stereoselectivities: (*2S*,*3S*,*S*)-**4an**, (*2S*,*3R*,*S*)-**4an**, (*2R*,*3R*,*S*)-**4an**, (*2R*,*3S*,*S*)-**4an**, (*2S*,*3S*,*R*)-**4am**, (*2S*,*3R*,*R*)-**4am**, (*2R*,*3R*,*R*)-**4am**, and (*2R*,*3S*,*R*)-**4am**. Therefore, the stereodivergent system was well catalyst-controlled during the formation of these isomers and was less affected by local stereochemical information from the substrates.

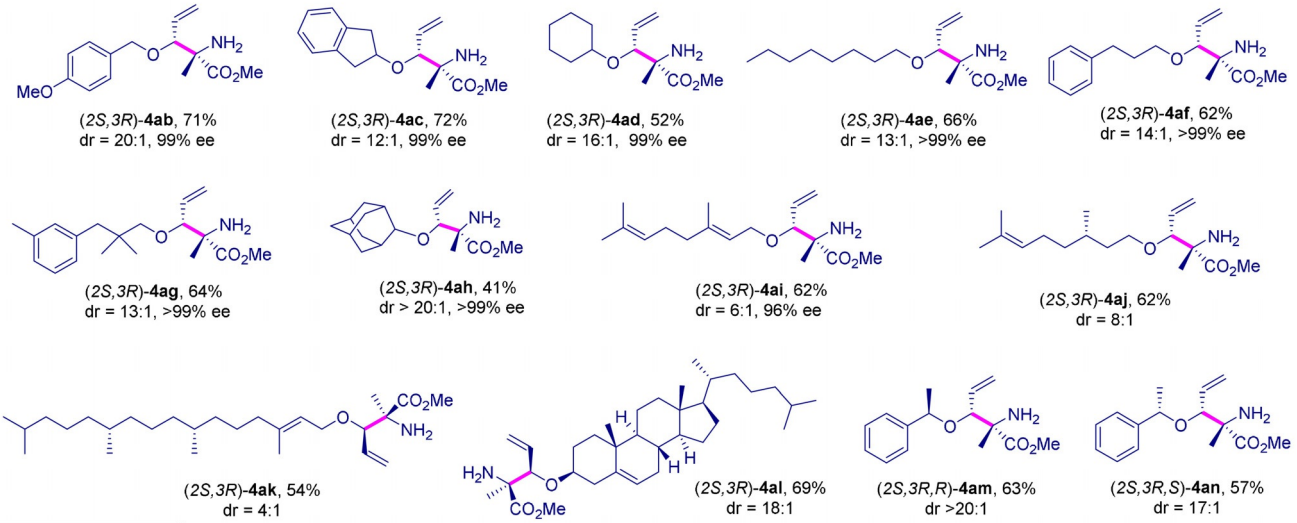
Using the reaction of **1a** and **2a** as a model, the stereoselectivities of this coupling reaction was rationalized in Scheme 4. Based on the previous report, the (S,S_p)-**L1**-Cu would activate **1a** under basic conditions to form metalated azomethine ylide **A**, which serves as a nucleophile for the coupling.[26] Because the *Re*-face of **A** is blocked by the bulky ^{i}Pr group of the oxazoline ring, the coupling occurs from the *Si*-face to give *2S* configuration in the product **3aa**. On the other hand, the Pd-allyl serves as an electrophile, which has two possible conformers **B** and **C**. Because of the stabilization of the allyl cation by the BnO group, the allylic substitution prefers branched regioselectivity. In the conformer **C**, the OBn group is closed to the bulky ^{t}Bu group of the ligand, thus the steric hindrance makes this conformer less favored than conformer **B**. Nucleophile-electrophile cross-coupling of **A**

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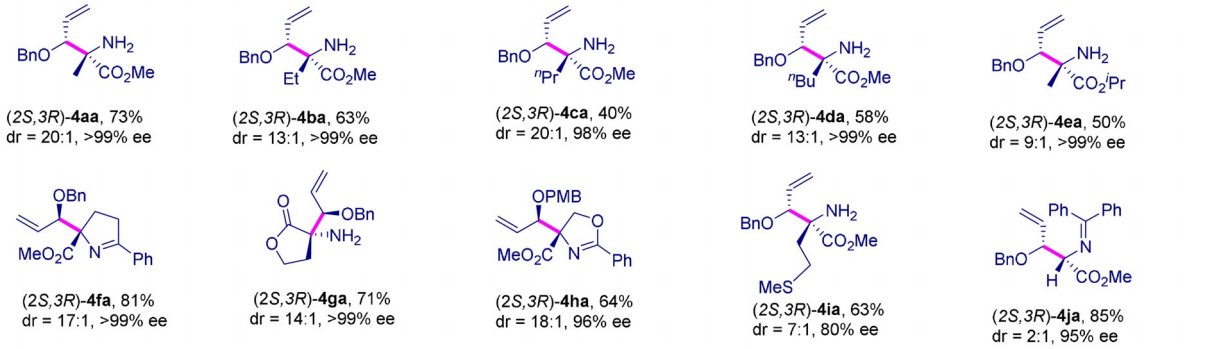
Table 3: Substrate scope of the *syn*-selective coupling reaction.[a]



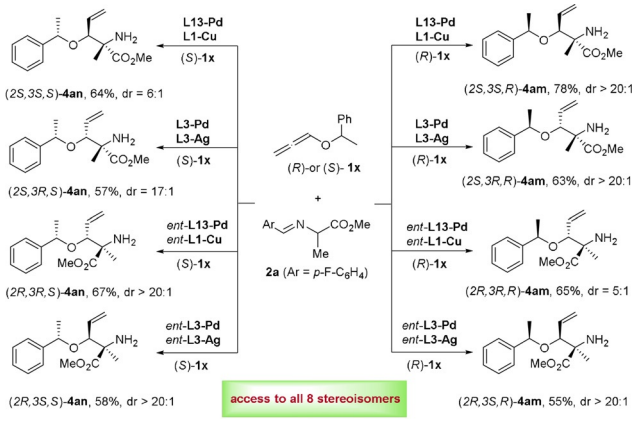
Alkoxylallene scope



Aldimine ester scope



[a] Reaction conditions: (i) **1a** (0.1 mmol), **2a** (0.2 mmol), **L13**-Pd(allyl)BF_4 (4 mol%), **L3** (5 mol%), AgOAc (6 mol%), Et_3N (200 mol%), CPME (0.5 mL), 30 °C, 48 h. (ii) Aqueous citric acid (10%), 1 h. The regioselectivity was >20:1 in all cases. The dr values were determined from the ^1H NMR spectra of the crude products. Isolated yields are given. The *ee* values were determined by chiral HPLC.



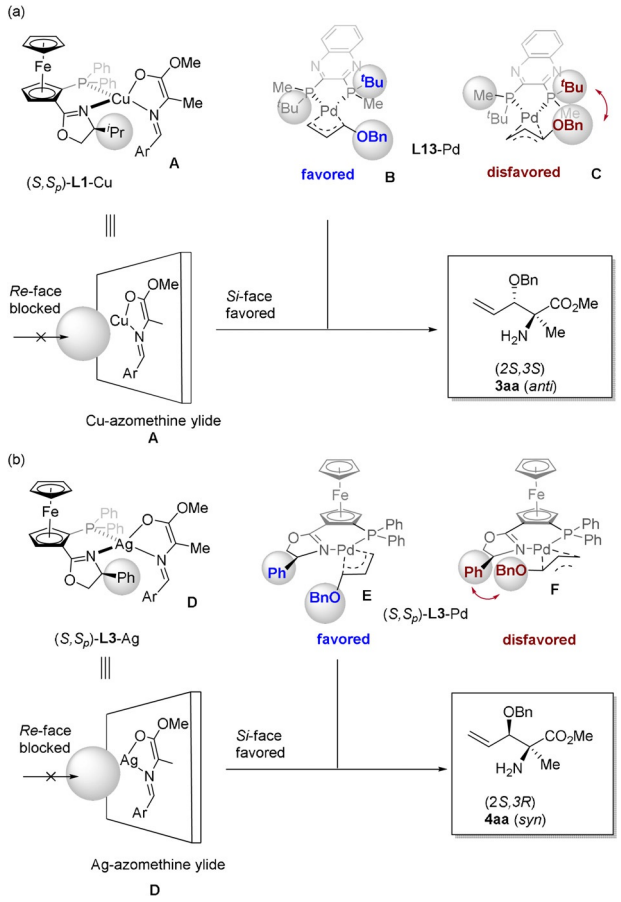
Scheme 3. Stereodivergent synthesis of products with three stereocenters.

and **B** leads to **3aa** with a stereochemistry of (*2S,3S*) (Scheme 4a).

The (S,S_p)-**L3**-Ag derived Ag-azomethine ylide **D** has a similar stereo-induction. Because the phenyl group also shield the *Re*-face and the coupling takes place from the *Si*-face to give *2S* configuration in the product **4aa**. For the (*S,S_p*)-**L3**-Pd-allyl species, due to *trans* effect of phosphorus atom, the allylic substitution prefers happening at the allyl terminus *trans* to phosphorus.[27] Among the two possible Pd-allyl species **E** and **F**, the latter has a strong repulsion between the BnO group and Ph group. Therefore, Nucleophile-electrophile cross-coupling occurs between **E** and **F** leading to **4aa** with a stereochemistry of (*2S,3R*) (Scheme 4b).[28]

The coupling products were rich in functional groups that could serve as handles for further derivatization (Scheme 5). For example, protection of the amine of (*2S,3S*)-**3aa** with

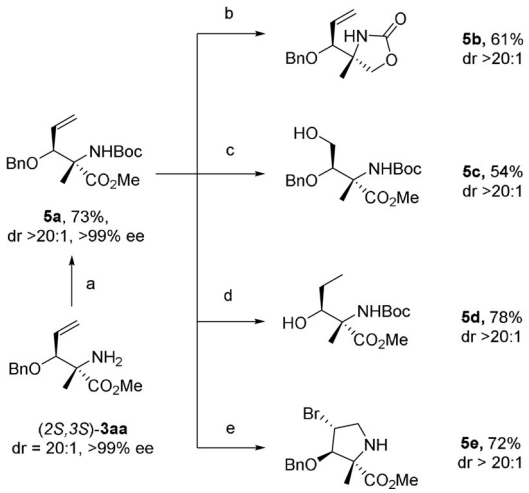
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Scheme 4. Rationalization of the stereoselectivities.

a *tert*-butoxycarbonyl group provided **5a** (73% yield), the ester group of which could be reduced to a hydroxyl group to give **5b** in 61% yield. The double bond of **5a** underwent ozonolysis, and subsequent reduction with $NaBH_4$ furnished **5c** in 54% yield. In addition, under hydrogenation conditions, the Bn group of **5a** could be removed, and then the alkene could be transformed to an ethyl group to give **5d** in 78% yield. Finally, upon treatment of **5a** with *N*-bromosuccinimide, a bromocyclization reaction occurred, giving tetrahydropyrrole **5e** with good diastereoselectivity.

To further demonstrate the utility of this stereodivergent coupling method, we used it for total syntheses of two amino alcohol natural products, mycestericins F and G (Scheme 6).[29] First, we carried out a scaled-up coupling reaction of *para*-methoxybenzyl-protected alkoxyallene **1a** and oxazoline **2h** by using the **L13**-Pd/**L1**-Cu system, which afforded corresponding *anti*-coupling product (2*S*,3*S*)-**3ha** in 60% yield with 99% *ee* and 6:1 dr. Removal the protecting group afforded alcohol **6a**, which was subjected to an olefin metathesis reaction with ketoalkene **7** and Hoveyda-Grubbs II catalyst to give **8a** in 65% yield. Hydrogenation of the double bond and hydroxylation of the oxazoline ring and ester group completed the total synthesis of mycestericin F. Using a similar route, but starting from *syn*-coupling product (2*S*,3*R*)-**3hb** obtained with the **L3**-Pd/**L3**-Ag catalyst system, we also achieved a total synthesis of mycestericin G. The



Scheme 5. Synthetic transformations of a coupling product. Reaction conditions: (a) $(Boc)_2O$, Et_3N, THF, 0°C to rt, 15 h. (b) $LiAlH_4$, THF, 0°C to rt, 12 h. (c) O_3 then $NaBH_4$, 3:1 dichloromethane/MeOH, −78°C to rt, 2 h. (d) Pd/C (10 mol%), H_2 (1 atm balloon), MeOH, rt, 5 h. (e) *N*-bromosuccinimide, Na_2CO_3, MeCN, rt, 15 h. Boc = *tert*-butoxycarbonyl.

spectra of the synthetic samples were identical to the reported spectra.[30]

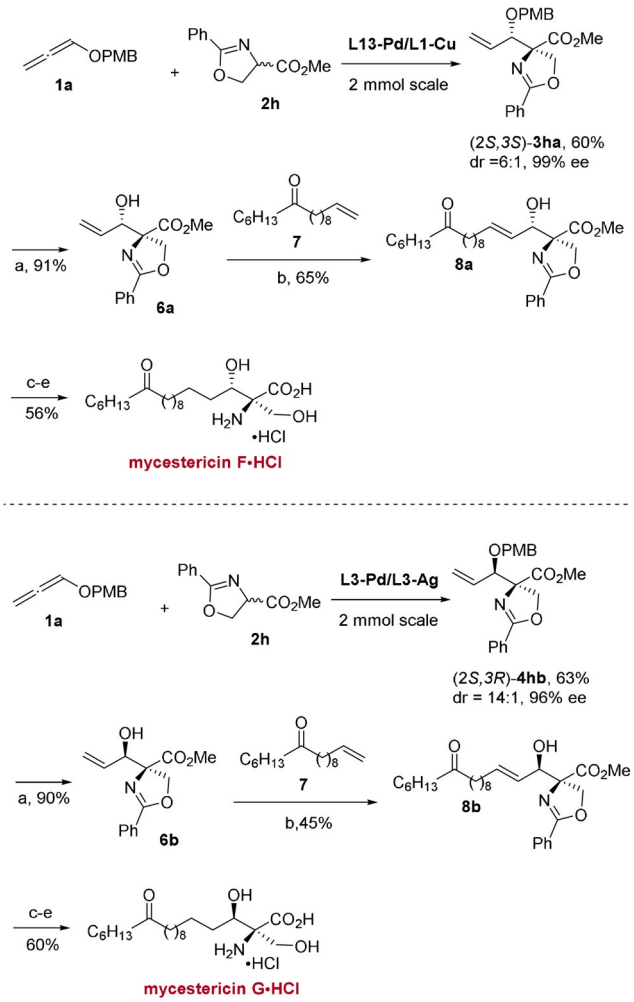
Conclusion

In summary, we have developed a protocol for diastereodivergent coupling of alkoxyallenes with aldimine esters. By choosing the appropriate dual-metal catalyst system (Pd/Cu or Pd/Ag) and appropriate chiral ligands, we could synthesize both *anti*- and *syn*-β-amino alcohol motifs with high enantioselectivity and diastereoselectivity from the same set of starting materials. These mild reactions have a broad substrate scope and excellent functional group tolerance. The coupling products could be further manipulated to access various useful synthons. Stereodivergent syntheses of all four stereoisomers of β-amino alcohols with vicinal stereocenters were achieved simply by varying the configurations of the two chiral metal catalysts. Using this coupling reaction as the key step, we accomplished total syntheses of natural products mycestericin F and G.

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Scheme 6. Total syntheses of mycestericins F and G: (a) 2,3-dicyano-5,6-dichlorobenzoquinone, CH_2Cl_2/H_2O, rt, 0.5 h. (b) Hoveyda-Grubbs II catalyst, benzene, 80 °C, 24 h. (c) Pd/C (10 mol%), H_2 (1 atm balloon), EtOAc, rt, 12 h. (d) Concentrated HCl in EtOH (6 M), reflux, 12 h. (e) NaOH (1 M), MeOH then HCl (1 M) workup. PMB = *para*-methoxybenzyl.

Conflict of interest

The authors declare no conflict of interest.

Keywords: amino alcohols · asymmetric catalysis · palladium · stereodivergence · synergistic catalysis

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